

Part 3 of 4

Precision-Guided Munitions: Infra-Red-Guided Weapons



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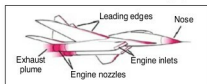


Fig. 1: Hot spots on the target aircraft as seen by an IR seeker

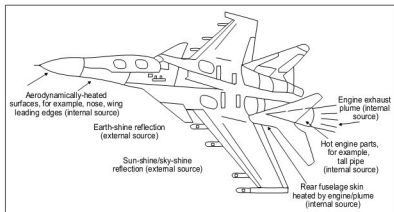


Fig. 2: Sources of IR signatures

Infra-red-guided [(IR)-guided] weapons are no less important than laser-guided munitions considering their usage in tactical warfare. While laser-guided munitions are predominantly used in air-to-surface and surface-to-surface roles, IR-guided weapons are largely surface-to-air man-portable air defence systems (MANPADS) and air-to-air missiles. The IR-guided air-to-air missile has been a feature of the fighter armament since 1950's and is likely to remain a key weapon for decades to come.

While laser and radar-guided weapons use semi-active or active homing for guidance, and therefore need an external laser source or radar for the purpose, IR-guided weapons make use of passive homing guidance in which the weapon homes on to the IR signatures due to hot areas of the target. For this reason, these are popularly known as heat-seeking weapons. Of course, for the weapon to be reliable, the guidance system processor has to perform the complex task of detecting and identifying the genuine signal in the presence of noise produced by unwanted IR emissions from

the background due to reflection of solar radiation from the Earth's surface, clouds and mountains, and IR countermeasures such as chaff and flares deployed by the target.

Initial development of IR-guided (or heat-seeking) missiles began in late 1950's. AIM-4 Falcons and AIM-9 Sidewinders are important air-to-air heat-seeking missiles of that time. These weapons were not very successful, initially, due to certain issues that were mainly related to reliability. Modern IR-guided missiles are a vast improvement over the ones developed in yesteryears. These not only have increased operational ranges and hit accuracy, these are far less vulnerable to countermeasures and are capable of discriminating the intended target from passive and active countermeasures.

IR homing guidance

There are two broad categories of IR-guided weapons, namely, those employing non-imaging-type IR-seeker heads and those employing imaging infra-red (IIR) seekers. Non-IIR-guided missiles home on to IR signatures produced by hot areas of the target. Fig. 1 shows hot spots on the target aircraft as seen by the seeker head of an IR-guided missile.

Fig. 2 shows different sources of IR signatures of the aircraft. The IR sensor used in this case is a single detector. In the case of IIR seekers, it is an imaging IR sensor, which is a focal plane array of IR/UV detectors. Imaging sensor array sees in IR in a manner similar to the functioning of a CCD sensor in a webcam or a digital camera.

IIR sensor output requires much more complex signal processing. In addition to being more resistant to IR countermeasures, such as flares and decoys, imaging seekers are also less likely to be fooled into locking onto the sun, another common trick for avoiding heat-seeking missiles. By using advanced image processing tech-